

Claude Code Master Integration Prompt

Integrate the Brisket Dynasty Valuation Model into Risk It to Get the Brisket

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You are acting as the senior quantitative engineer, fantasy-football modeling lead, data architect, backend architect, product systems architect, and implementation coordinator for the Risk It to Get the Brisket platform.

This is an **implementation assignment**, not another open-ended research assignment and not a request for a conceptual response only.

Your job is to inspect the existing repository, reconcile the current production architecture with the combined dynasty-model research summarized below, and begin integrating a production-quality dynasty valuation system covering offense, IDP, rookie picks, market comparison, roster analysis, rest-of-season value, and trade optimization.

The attached research document is:

- BDVM v1 complete.pdf
- Full title: **The Brisket Dynasty Valuation Model (BDVM) v1.0**
- It includes research/design Parts 1–14 and executable Appendices A–F:
 - dynasty_engine.py
 - run_examples.py
 - examples_output.txt
 - schema.sql
 - league_config.example.json
 - player_payload.example.json

Treat that PDF as a critical implementation specification, but **do not blindly paste its reference implementation into production**. The appendices are an executable mathematical reference and acceptance fixture. The existing repository, actual data pipelines, current APIs, current UI contracts, current agents, and current production conventions must be audited before integration.

The combined conclusion from both independent research efforts is:

Build a transparent, modular, projection-driven dynasty model first. Use dynamic replacement value, position-specific conditional aging, separate fantasy-relevance survival, role ascension, probabilistic uncertainty, and strategy-specific time utility. Keep the independent fundamental model separate from the market. Use actual roster marginal utility for personalized roster and trade decisions. Introduce machine-learning challengers only after clean historical snapshots and rolling backtests exist.

1. Mission

Build and integrate a Version 1 dynasty valuation engine that can accept replaceable upcoming-season projections and produce, for every offensive and defensive player:

- upcoming-season value;
- rest-of-season value;
- independent fundamental dynasty value;
- contender value;

- balanced value;
- rebuild value;
- market value;
- model-versus-market gap;
- market-adjusted value;
- position-adjusted cross-position rank;
- trade units;
- floor, median, and ceiling;
- confidence and volatility;
- probability of remaining fantasy-startable;
- breakout, bust, elite-asset, and one-year value-collapse probabilities;
- a multi-year value path;
- a plain-English explanation of the largest value drivers.

The model must support:

- QB;
- RB;
- WR;
- TE;
- DT/interior defensive line;
- DE/EDGE;
- off-ball LB;
- CB;
- S;
- combined DL;
- combined DB;
- platform-specific hybrid eligibility;
- 1QB;
- Superflex;
- 2QB;
- TE premium;
- arbitrary PPR and points-per-first-down rules;
- arbitrary IDP scoring;
- position-specific IDP scoring;
- shallow and deep lineups;
- best-ball and managed-lineup leagues;
- configurable roster and waiver depth;
- rookie picks;
- future roster analysis and trade optimization.

This engine must become a shared foundation for the existing:

- dynasty rankings;
- trade calculator;
- roster analyzer;
- trade finder;
- rest-of-season rankings;
- contender/rebuilder tools;
- player cards;
- player career-value charts;
- buy-low and sell-high tools;
- league-specific values;
- IDP tools.

Do not create a conflicting second valuation universe beside the existing system. Audit what exists, define migration/adaptation boundaries, and integrate through stable interfaces and versioned snapshots.

2. Source priority and conflict resolution

Use this priority order:

1. **Actual repository and runtime truth**
 - Current source files, live routes, real schemas, active jobs, current data contracts, and working tests are authoritative about what exists.
 - Compiled artifacts, stale docs, abandoned branches, and old generated output are not authoritative.
2. **The architectural decisions in this prompt**
 - These are the chosen direction unless a direct repository constraint or an actual backtest proves that a specific decision is wrong.
3. **BDVM v1 complete.pdf**
 - Use its formulas, priors, data dictionary, examples, schema concepts, and appendices as the mathematical reference.
 - Preserve its labels distinguishing established evidence, recommendations, hypotheses, and unvalidated priors.
4. **Existing project docs and previous prompts**
 - Reuse valid work, but reconcile it with current source reality.
5. **Your own judgment**
 - Use it where evidence or requirements remain incomplete.
 - Document every material judgment call.
 - Do not silently invent unsupported formulas.
 - Do not stop the entire implementation because an advanced field is unavailable. Use a documented proxy or omit the feature while reducing confidence.

When two design approaches conflict, prefer the approach that best balances:

- out-of-time predictive accuracy;
 - explainability;
 - maintainability;
 - data availability;
 - IDP sample size;
 - leakage prevention;
 - compatibility with the existing platform.
-

3. Non-negotiable model decisions

3.1 Keep value concepts separate

The system must maintain at least these distinct concepts:

Independent fundamental dynasty value

Calculated without reading current dynasty market values.

It may use:

- projection consensus;
- league scoring;
- dynamic replacement;
- position and archetype;
- age trajectory;
- career mileage;
- role ascension;
- survival/startability;
- expected games;
- injury-specific durability;
- contract and team investment for future role security;

- uncertainty;
- time discounting;
- strategy-specific usability.

It must be capable of disagreeing with the market.

Market value

Estimated independently from:

- crowdsourced preference;
- revealed trades;
- expert consensus;
- startup ADP;
- rookie ADP;
- market momentum;
- source dispersion;
- liquidity.

Market-adjusted value

An optional blend used for market-aware ranking or expected trade-clearing estimates.

Rest-of-season value

A single-horizon, remaining-week value. It is not dynasty value and is not market value.

Roster-specific marginal value

The change in one actual roster's optimized lineup output, resilience, playoff odds, championship odds, or long-term utility if the player is added or removed.

Contender, balanced, and rebuild value

Three strategic currencies created inside the future-season sum. They must not be implemented as a post-hoc "young/old" multiplier on one universal value.

Never overwrite one value concept with another. Store and expose them separately.

3.2 Use the BDVM central formula as the Version 1 skeleton

For player i , strategy s , and future season t :

$$DV(i, s) = \sum_{t=0}^{H_s} u_s(t) d_s^t S_i(t) E[\max(0, \mu_i(t) - R_{p,\ell}(t))] G_i(t) - \lambda_s \Psi_i$$

Where:

- $\mu_i(t)$: expected fantasy points per game, conditional on playing and retaining a role;
- $R_{p,\ell}(t)$: dynamic league- and eligibility-specific replacement FPG;
- $E[\max(0, \mu - R)]$: expected positive starter surplus under the outcome distribution;
- $G_i(t)$: expected games played;
- $S_i(t)$: probability the player remains fantasy-startable in season t ;
- d_s : strategy-specific annual discount factor;
- $u_s(t)$: strategy-specific usability of production in year t ;
- $\lambda_s \Psi_i$: configurable certainty-equivalent risk preference adjustment;
- H_s : strategy horizon.

This is the player-level fundamental skeleton.

The roster analyzer and personalized trade evaluator must additionally calculate actual before/after roster utility rather than assuming displayed trade values are additive.

3.3 Use expected positive surplus, not only a plain difference

The Version 1 player-level surplus is:

$$SSV_i(t) = E[\max(0, X_i(t) - R_{p,\ell}(t))] \cdot G_i(t)$$

When $X_i(t) \sim \text{Normal}(\mu_i(t), \sigma_i(t))$:

$$E[\max(0, X - R)] = (\mu - R)\Phi(z) + \sigma\phi(z)$$

$$z = \frac{\mu - R}{\sigma}$$

This prices a player as an option on startable production because a fantasy manager is not forced to start a below-replacement outcome.

Important implications:

- upside can create value for rookies and uncertain players without an arbitrary youth bonus;
- volatile CB and sack-dependent EDGE profiles can be handled with the same mathematics;
- volatility should not automatically be treated as entirely bad;
- players far above replacement are affected less by this option effect;
- roster-spot costs and actual roster marginal utility must prevent the system from overvaluing endless volatile bench players.

This is a **central Version 1 hypothesis, not an untouchable law**. Build an explicit ablation against:

$$\max(0, \mu - R)$$

and, where appropriate:

$$\mu - R$$

The option-value formulation remains the champion unless rolling out-of-sample testing shows that a simpler formulation performs as well or better.

3.4 Separate conditional aging from survival

Never merge these questions:

1. How productive is the player if he remains in a meaningful role?
2. What is the probability he retains a meaningful role?

Implement:

Conditional production/trajectory

$$E[\mu_i(t) \mid \text{startable}]$$

Fantasy-relevance survival

$$S_i(t) = \prod_{j=1}^t (1 - h_i(j))$$

Aging curves must be fitted on consecutive player-year pairs with meaningful participation in both years or another method that controls survivorship.

Loss of role, benching, retirement, replacement, and disappearance belong in the hazard/survival model, not inside the conditional aging curve.

This separation is mandatory because combining them causes double-counting when role, contract, injury, and age penalties are later added.

3.5 Include a distinct role-ascension module

A projection-only future model systematically undervalues:

- rookies;
- second-year players;
- young TEs;
- players whose current route/snap share leaves an identifiable growth path;
- highly drafted players not yet in full roles.

Use a bounded ascension term:

$$\mu_i(t) = \mu_i(0) \frac{A_p(a_{eff}(t))}{A_p(a_{eff}(0))} [1 + \kappa_i(1 - e^{-t/1.5})]$$

Where:

- A_p is the position/archetype age curve;
- a_{eff} is age adjusted for career mileage;
- κ_i is a capped role-ascension coefficient.

The ascension coefficient may use:

- draft capital with decay;
- college production with decay;
- current versus projected route/snap opportunity;
- competition;
- vacated opportunity;
- receiving-role indicators;
- prospect quality;
- recent NFL evidence.

It must not become a generic youth multiplier.

3.6 Replacement level performs the positional-scarcity work

Do not add arbitrary Superflex, TE-premium, EDGE, LB, or safety multipliers.

Format sensitivity must flow from:

- exact scoring;
- required lineup slots;
- flex and Superflex eligibility;
- IDP flex;
- position-specific requirements;

- roster depth;
- number of teams;
- waiver buffer;
- position eligibility;
- available player pool.

Version 1 neutral player value may use the BDVM rank-based method:

$$R_{p,\ell} = FPG \text{ at rank } [\text{league-wide flex-allocated startable slots} + \text{waiver buffer}]$$

Use VORP/waiver replacement as the primary baseline. Retain worst-starter/VOLS as a diagnostic.

Later or in parallel, implement a representative-roster or lineup-optimization challenger.

For actual roster analysis and trade evaluation, use the real roster's replacement alternatives and optimized legal lineup:

$$MU_{player,R} = OPT(R \cup player) - OPT(R \cup alternative)$$

This is the reconciliation between the two research approaches:

- **rank-based dynamic VORP is the transparent, fast, league-neutral Version 1 player engine;**
- **actual lineup optimization is the authoritative personalized roster and trade engine.**

Add monotonicity and sanity tests. Increasing starter demand or roster depth must move replacement and scarcity in a logically consistent direction. Do not accept a plausible-looking replacement table without these tests.

3.7 Do not double-count information already in the current projection

Encode this as a hard architectural rule:

Any variable likely incorporated into the current projection may affect future trajectory, ascension, survival, games, or uncertainty, but must not independently adjust $\mu(0)$ unless a confirmed event occurred after the projection timestamp and is not reflected in the source.

Examples:

Variable	Re-add directly to current projection?	Legitimate use
Team quality	No	Future persistence or projection staleness
Offensive/defensive scheme	No	Role stability, archetype, uncertainty
QB quality	No	Future persistence and scenario risk
Offensive line	No	Future uncertainty, not a current bonus
Current snap/route/target/carry share	No for current mean	Role security and uncertainty
Coaching change	No	Structured event, future stability
Known injury designation	No if projection is current	Games, ramp-up, durability
Age	Yes for future seasons	Trajectory
Draft capital	Limited	Ascension and survival with decay
College production	Rookie/Year 2 only	Ascension prior
Contract	No	Future role and survival
New event after source date	Yes, through event update	Only proportional to confidence and unreflected share

Every projection row must have an `as_of` timestamp. Every event must know whether it has already been reflected.

3.8 Prefer raw statistical projections

Accept both:

- raw projected statistics;
- direct fantasy-point projections.

Prefer raw statistics because the platform must calculate fantasy points internally under exact league scoring.

Every projection source adapter must record:

- source;
- snapshot timestamp;
- season;
- player identity;
- raw stats where available;
- direct projected fantasy points where available;
- projected games;
- projected opportunity;
- source floor/median/ceiling where available;
- source version;
- scoring assumptions if direct fantasy points are supplied.

Do not tie the dynasty model to one projection provider.

3.9 Treat IDP as a first-class domain

Do not implement offense first and leave IDP as a generic multiplier or external-value lookup.

The core model must support:

- DT/interior defensive line;
- EDGE/DE;
- off-ball LB;
- CB;
- S;
- combined DL;
- combined DB;
- platform eligibility;
- true-position taxonomy;
- designation-change scenarios.

IDP must participate in:

- projection ingestion;
 - exact scoring;
 - replacement;
 - aging;
 - survival;
 - uncertainty;
 - market comparison;
 - roster analysis;
 - trade search;
 - player explanations;
 - backtesting.
-

3.10 Version 1 must be transparent and parametric

Do not start with a single end-to-end gradient-boosted or neural model.

Version 1 should use:

- transparent formulas;
- position/archetype curves;
- bounded modifiers;
- survival/hazard modeling;
- probabilistic distributions;
- Monte Carlo where necessary;
- explainable decompositions;
- configuration-driven priors.

Machine learning may later replace targeted modules:

- role ascension;
- games missed;
- startable-role hazard;
- projection blending;
- market clearing;
- trade acceptance.

A challenger is promoted only when it beats the parametric champion on rolling unseen seasons and does not materially degrade:

- IDP;
 - rookies;
 - top assets;
 - calibration;
 - interpretability;
 - latency;
 - robustness.
-

3.11 The market runs after the fundamental model

The fundamental model must be computed and stored before any market values are read into the run.

Store:

- fundamental value;
- market value;
- market-adjusted value;
- market gap;
- liquidity;
- market dispersion;
- market momentum;
- source-type breakdown.

The market may not silently regularize the fundamental score.

Use the market as:

- a separate price;
 - a comparison layer;
 - an optional display blend;
 - a trade-clearing estimate;
 - a liquidity signal;
 - a prior only in a separately labeled market-adjusted output.
-

3.12 Do not naively sum incompatible offense and IDP market values

The current site may use:

- KTC as an offensive market anchor;
- IDP Trade Calculator as a defensive market anchor;
- additional consensus sources.

Audit the current normalization before changing anything.

Never call a mixed offense-IDP trade externally fair by directly adding raw KTC values to raw IDP Trade Calculator values unless a validated shared normalization already exists.

Store:

- `raw_market_value`;
- `raw_market_source`;
- `raw_market_type`;
- `normalized_market_value`;
- `normalization_version`;
- `normalization_confidence`.

The **fundamental engine naturally provides a common offense-IDP scale** through league scoring and replacement. The market layer still requires a defensible mapping to a common latent market scale.

3.13 Use two trade-evaluation layers

Generic package display

A configurable concentrated-package function may use the BDVM CES formulation:

$$PackageValue = \left(\sum_i v_i^\theta \right)^{1/\theta} - C_{spot} \cdot Overflow$$

Starting priors:

- $\theta = 1.20$;
- raise toward 1.35 in shallow leagues;
- $C_{spot} = 120$ on a 0–10,000 scale.

Personalized authoritative verdict

$$\Delta U = U(Roster_{after}) - U(Roster_{before})$$

Include:

- optimized lineup impact;
- best-ball inclusion;
- projected wins;
- playoff/championship delta;
- depth resilience;
- positional need;
- competitive timeline;
- roster-spot cost;
- liquidity;
- risk concentration;
- future picks;
- opponent utility;
- expected acceptance.

Do not judge all two-for-one trades by adding displayed player values.

4. Exact Version 1 mathematical specification

Use the following as the initial production hypothesis, with every constant stored in a versioned parameter set rather than buried in code.

4.1 Projection consensus

For source m , player i , statistic k , and position p :

$$\hat{x}_{i,k} = \sum_m w_{m,k,p} Winsorize(x_{i,k,m})$$

Recommended initial behavior:

- if honest source-accuracy history is unavailable, use equal weights;
- start archiving all source snapshots immediately;
- use weighted trimmed mean where at least five credible sources exist;
- otherwise use weighted mean plus weighted median robustness check;
- cap one source at 35%;
- shrink learned weights toward equal weighting;
- calculate weights by position and, where data permits, by statistic and player tier;
- train weights only on completed prior seasons;
- penalize stale sources;
- store source disagreement as uncertainty.

Suggested shrinkage:

$$w_m = 0.30 \left(\frac{1}{M} \right) + 0.70 \tilde{w}_m$$

Do not fake historical weights using current revisions of past projections.

4.2 League scoring

$$FP_{i,\ell} = \sum_k c_{\ell,k} \hat{x}_{i,k} + B_{\ell}(\hat{\mathbf{x}}_i)$$

Support:

- linear scoring;
- stacked scoring categories;
- nonlinear thresholds;
- bonuses;
- penalties;
- TE premium;
- points per first down;
- return yards;
- position-specific IDP scoring;
- six-point passing TDs;
- arbitrary supported league rules.

Scoring configuration must be data-driven.

4.3 Conditional age curve

Initial parametric family:

$$A_p(a) = \begin{cases} \exp[-c_{up,p}(peak_p - a)^2], & a < peak_p \\ \exp[-c_{dn,p}(a - peak_p)^2], & a \geq peak_p \end{cases}$$

Use continuous archetype blends where appropriate, especially pocket versus dual-threat QB.

Starting priors from BDVM:

Group	Peak	c_{up}	c_{dn}
QB pocket	30.0	0.006	0.010
QB dual	26.0	0.010	0.028
RB	25.0	0.012	0.030
WR	27.0	0.014	0.024
TE	27.5	0.030	0.018
DT	27.5	0.016	0.020
EDGE	27.0	0.014	0.022
LB	26.5	0.012	0.030
CB	26.0	0.010	0.030
S	27.0	0.012	0.026

These are priors, not proven truths.

4.4 Career mileage

$$a_{eff} = a + \min(Cap_p, \max(0, (Load - Reference_p)/PerYear_p))$$

Starting priors:

Position	Load	Reference	Per effective year	Cap
RB	Career touches	750	500	2.5 years
WR	Career targets	500	450	1.5
TE	Career targets	400	450	1.5
QB	Career dropbacks	2,000	2,500	1.0
LB/EDGE/CB/S	Defensive snaps	3,000	2,200–2,400	1.5
DT	Defensive snaps	3,000	2,400	1.0

Test whether workload remains predictive after age, role, and quality controls. Do not treat mileage as a guaranteed RB cliff.

4.5 Draft and college decay

Draft-capital weight:

$$w_{draft}(n) = \exp[-0.45(n - 1)]$$

College weight:

$$w_{college}(n) = \exp[-1.10(n - 1)]$$

Where $n = 1$ is rookie season.

Use meaningful-NFL-sample gates as an additional decay mechanism:

- WR/TE/RB receiving: routes;
- RB: carries plus targets;
- QB: dropbacks;
- IDP: defensive snaps or role-specific snaps.

College and athletic data should be effectively absent for established veterans.

4.6 Future mean

$$\mu_i(t) = \mu_i(0) \frac{A_p(a_{eff,i} + t)}{A_p(a_{eff,i})} [1 + \kappa_i(1 - e^{-t/1.5})]$$

For future seasons, regress unstable event conversion:

- offensive TD rate;
- sack conversion;
- interception rate;
- defensive touchdown rate;
- forced-fumble rate.

Starting priors:

- offensive TD regression: 0.40 toward expected/baseline;
- EDGE/DT sack-to-expected-sack regression: 0.35;
- apply to $t \geq 1$, not as an arbitrary current-season haircut when the current projection already incorporates regression.

4.7 Uncertainty

Initial structure:

$$\sigma_i(t) = \sqrt{[CV_p(1 + 0.9RoleUncertainty_i)(1 + 0.45EventVolatility_i)(1 + 0.6SmallSample_i)\mu_i(t)]^2 + \sigma_{source,i}^2 + [\psi_p\mu_i(t)\sqrt{t}]^2}$$

Starting priors:

Position	Base CV	Future drift ψ
QB	0.26	0.16
RB	0.36	0.30
WR	0.34	0.22
TE	0.36	0.22
DT	0.32	0.20
EDGE	0.34	0.22
LB	0.26	0.22
CB	0.40	0.30
S	0.30	0.24

Decompose uncertainty into user-facing components:

- projection-source;
- role;
- injury;
- event dependence;
- small sample;
- contract;
- eligibility;
- future drift.

4.8 Survival

Use a discrete-time hazard model:

$$\text{logit}(h_{i,t}) = \alpha_{p,t} + f_p(\text{age}_{i,t}) + g_p(\text{experience}_{i,t}) + \beta_p X_{i,t}$$

Inputs may include:

- current and recent snap share;
- routes/pass-rush snaps/coverage role;
- contract security;
- guaranteed money;
- practical exit year;
- recent extension;
- competition;
- draft capital times decay;
- recency-weighted games missed;
- same-type recurrence;
- time since soft-tissue injury;
- archetype;
- eligibility risk;
- recent role trend.

Starting base annual hazard priors from BDVM should be stored as a versioned table, not hardcoded. Fit the first production version to the platform’s own definition of “fantasy-startable” as soon as historical data allows.

Primary calibration target:

Average predicted $S(3)$ by position among players startable at Year 0 must match observed three-year startable retention.

4.9 Expected games

Model games separately from fantasy-relevance survival.

Use:

- current projected games for $G(0)$;
- recency-weighted games missed;
- injury type and recurrence;
- expected return date;
- workload ramp;
- position;
- age;
- suspension only when confirmed;
- uncertainty bands.

Do not use a raw career injury count.

4.10 Strategy profiles

Starting priors:

Profile	Discount d	Horizon	Usability $u(0..)$	Risk λ	Pick premium
Contender	0.72	4	1.00, 0.95, 0.80, 0.65, 0.55	+0.20	0.92

Profile	Discount d	Horizon	Usability $u(0..)$	Risk λ	Pick premium
Balanced	0.85	6	1.00, 1.00, 1.00, 0.95, 0.90, 0.85, 0.80	+0.10	1.00
Rebuilder	0.93	8	0.45, 0.75, 1.00, 1.00, 1.00, 0.95, 0.90, 0.85, 0.80	-0.10	1.08

Also calculate a risk-neutral expected-value output with $\lambda = 0$.

Do not present the negative rebuilder risk coefficient as empirically established. Keep it configurable and run sensitivity tests because option value is already present in $E[\max(0, X - R)]$.

4.11 Fundamental trade scale

Starting conversion:

$$TV_{i,s} = 10000 \left(\frac{DV_{i,s}}{Anchor_s} \right)^\gamma$$

Starting $\gamma = 1.20$.

Use a stable calibration anchor and preserve the ability to change display scaling without changing the underlying present value.

Also expose a 0–100 percentile score.

4.12 Market gap

Normalize market sources to a comparable latent scale before aggregation.

$$Gap_i = TV_{i,balanced} - TV_{i,market}$$

$$Alpha_i = Gap_i \cdot Liquidity_i$$

Store both raw gap and confidence/liquidity-adjusted gap.

Optional market-adjusted value:

$$MA_i = FUND_i + \lambda_M (MARKET_i - FUND_i)$$

Starting outputs:

- independent fundamental: $\lambda_M = 0$;
- market-adjusted player display: $\lambda_M = 0.25$;
- trade-clearing estimate: approximately $\lambda_M = 0.50$, subject to validation.

4.13 Rest of season

Use:

$$\mu_{ROS} = w_{proj}\mu_{preseason,updated} + (1 - w_{proj})\mu_{recent}$$

$$w_{proj} = \frac{n_{prior}}{n_{prior} + weeks\ played}$$

Starting priors:

- offense $n_{prior} = 6$;
- IDP $n_{prior} = 8$.

$$ROS_i = \sum_{w=now}^{end} 1\{not\ bye\}P(active_{i,w})E[\max(0, X_{i,w} - R_{p,w})]PlayoffWeights(w)$$

Use:

- exact remaining schedule;
- bye;
- expected return;
- ramp-up;
- current role;
- trade deadline;
- league playoffs;
- actual roster bench baseline for personalized contender ROS.

ROS has no multi-year age or survival term.

5. Position-specific requirements

5.1 Quarterback

Maintain a continuous pocket/dual archetype blend.

Candidate classification:

- projected rushing points share;
- designed-run rate;
- scramble rate.

Key features:

- dropbacks;
- passing efficiency;
- designed rushing;
- scramble volume;
- goal-line rushing;
- sack rate;
- projected starts;
- contract security;
- recent draft investment at QB;
- backup quality;
- age;
- archetype.

Starting-job security is a major survival/role variable.

Do not add a Superflex multiplier. Superflex value must emerge from scoring, slots, replacement, and roster utility.

Add tests demonstrating sensible behavior when:

- moving from 1QB to Superflex;
- changing team count;
- increasing QB roster depth;
- changing six-point passing TD scoring.

5.2 Running back

Prioritize:

- carries;
- targets;
- routes;
- route participation;
- goal-line share;
- two-minute role;
- snap share;
- receiving role;
- projected games.

Future role/survival:

- age;
- career touches;
- recent workload;
- draft capital with decay;
- contract;
- team investment;
- competition;
- injury-specific recurrence.

Receiving opportunity should provide greater insulation than long-run efficiency.

Do not use career yards per carry as a standalone talent score.

Add a sell-window flag only as a labeled heuristic and only after comparing market and model. Do not let it alter the fundamental value outside the model.

5.3 Wide receiver

Prioritize:

- routes;
- route share;
- targets;
- target share;
- targets per route;
- yards per route;
- air-yard share;
- first-read share where available;
- red-zone opportunity;
- projected team passes.

Use overlapping metrics carefully through regularization, residualization, or feature selection.

Team change should increase uncertainty and reduce the reliability of prior efficiency.

Ascension is especially important for young WRs, but must remain bounded by actual opportunity and prospect evidence.

5.4 Tight end

Separate receiving TEs from block-heavy roles.

Prioritize:

- routes per team dropback;
- route participation;
- target rate;
- slot/wide usage;
- red-zone and end-zone usage;
- receiving/blocking split;
- age;
- draft capital with decay.

TE premium must be handled in scoring and replacement, not by a post-hoc TE multiplier.

A low route-share TE should carry high role uncertainty.

5.5 Interior defensive line

Prioritize:

- defensive snaps;
- pass-rush snaps;
- pressure opportunity;
- pressure rate;
- expected sacks;
- tackles and tackle opportunity;
- run-defense role;
- rotation.

DT-required and combined-DL formats must produce materially different replacement values.

5.6 EDGE/DE

Prioritize:

- pass-rush snaps;
- pressure rate;
- expected sacks;
- QB hits;
- hurries;
- three-down role;
- alignment;
- snap share;
- rotation;
- contract;
- age.

Sacks, forced fumbles, and defensive touchdowns are event-dependent. Increase uncertainty and regress future conversion.

Position eligibility is a first-class scenario.

5.7 Off-ball linebacker

Prioritize, in order:

- snap share;
- three-down role;
- coverage role;
- green-dot/play-caller role;

- tackle opportunity;
- tackle rate versus expected;
- box alignment;
- competition;
- scheme stability;
- contract.

LB generally has a stable tackle floor but can have a high role-loss hazard. Keep conditional production and role hazard separate.

High raw LB points must not override a high replacement level in deep tackle-heavy LB pools.

5.8 Cornerback

Use a separate CB model.

Do not use real-life coverage grade as a direct positive fantasy-production input.

Prioritize:

- coverage snaps;
- targets faced per coverage snap;
- slot share;
- run-support role;
- tackle opportunity;
- pass breakups;
- team pass volume faced;
- starting-role security.

Regress:

- interceptions;
- defensive touchdowns;
- forced fumbles.

Account for the possibility that elite shutdown quality suppresses fantasy opportunity.

Model rookie-CB target opportunity and its possible decay.

5.9 Safety

Classify continuously or probabilistically among:

- box;
- big nickel;
- slot;
- split field;
- single high;
- hybrid LB/S.

Prioritize:

- box share;
- slot share;
- deep share;
- tackle opportunity;
- blitz rate;
- run-fit role;
- full-time snaps;
- eligibility.

A box-to-deep role move must affect mean, uncertainty, and downside probability through a structured role event.

5.10 Position designation

Store both:

- true football position;
- platform position/eligibility.

For uncertain future designation:

$$Value_i = \sum_e P(e)Value_i(e)$$

Allow human review of probability assumptions.

6. Data architecture

6.1 Required conceptual tables

Adapt names to repository conventions, but support equivalent entities.

Identity and league

- players;
- player aliases/crosswalks;
- teams;
- league configs;
- scoring rules;
- lineup slots;
- position eligibility history;
- platform position history.

Projections

- projection sources;
- projection snapshots;
- player projection rows;
- projected stat rows;
- source accuracy;
- consensus runs.

Historical performance

- player week stats;
- player season stats;
- snap history;
- route/opportunity history;
- alignment history;
- advanced metrics;
- team environment snapshots.

Career and context

- draft records;
- college profiles;
- combine/athletic testing;
- injury events;
- contracts;
- contract years;

- depth-chart snapshots;
- coaching/scheme events;
- transactions;
- structured news events.

IDP

- true-position mappings;
- defensive role history;
- pass-rush metrics;
- tackle opportunity;
- coverage opportunity;
- alignment splits;
- eligibility scenarios.

Market

- market sources;
- source types;
- market snapshots;
- player market values;
- observed trades;
- startup ADP;
- rookie ADP;
- source dispersion;
- normalization mappings.

Modeling

- feature snapshots;
- model versions;
- parameter sets;
- model runs;
- projection consensus runs;
- replacement runs;
- age-curve versions;
- survival versions;
- calibration reports;
- player outputs;
- explanation components;
- pick distributions.

Roster and trades

- fantasy leagues;
- league snapshots;
- rosters;
- roster assets;
- future picks;
- roster analysis snapshots;
- positional profiles;
- competitive-window snapshots;
- trade candidates;
- trade packages;
- trade evaluations;
- acceptance estimates;
- recommendation reasons;
- review flags.

6.2 As-of snapshots are mandatory

Every historical or current model input must include:

- source timestamp;
- effective timestamp;
- ingestion timestamp;
- season/week;
- version;
- data-quality status.

Never overwrite historical source values.

Every model output must retain:

- model version;
- parameter-set ID;
- feature-schema version;
- projection snapshot;
- market snapshot;
- league config hash;
- code commit;
- random seed where applicable;
- training cutoff;
- as-of timestamp.

6.3 Missing-data behavior

Rules:

1. Never silently convert missing to zero.
2. Preserve missingness indicators.
3. Use position/experience priors only for low-impact optional fields.
4. If a mandatory projection is missing, fail or use a clearly labeled fallback.
5. Reduce confidence for missing:
 - source coverage;
 - contract;
 - injury;
 - role;
 - eligibility;
 - alignment.
6. Do not present imputed proprietary metrics as observed.
7. Use public-data fallback models when premium data are absent.
8. Preserve the last known good published snapshot if a new run fails.
9. Mark stale output visibly.

6.4 Explicitly excluded Version 1 inputs

Do not add these unless backtesting supports them:

- raw team win totals;
- distant strength of schedule outside ROS;
- subjective coaching quality scores;
- raw career injury count;
- combine athleticism for established veterans;
- unstructured social sentiment;
- arbitrary film grades;
- current dynasty rank inside fundamental value;
- generic position-agnostic age multipliers;
- arbitrary offense/defense market additions;
- future information in historical tests.

7. Structured event system

Create a closed event ontology.

Candidate event types:

- injury;
- surgery;
- practice limitation;
- activated/return;
- suspension;
- trade;
- signing;
- release;
- contract extension;
- fifth-year option;
- franchise tag;
- draft competition;
- depth-chart promotion;
- depth-chart demotion;
- coordinator change;
- scheme/role change;
- position-eligibility change.

Each event must store:

- event type;
- player/team;
- source;
- source reliability;
- publication timestamp;
- effective timestamp;
- expected impact target;
- confidence;
- expected duration;
- decay/half-life;
- whether projections already reflect it;
- human review status.

Generic structure:

$$Adjustment_e = Impact_e \cdot Confidence_e \cdot Reliability_e \cdot (1 - Reflected_e) \cdot 2^{-Days/HalfLife_e}$$

An event must update the relevant module, not directly add an arbitrary final-score bonus.

Examples:

- injury updates games, workload ramp, and durability;
 - drafted competition updates role and survival;
 - extension updates contract security;
 - alignment change updates position-specific mean and uncertainty;
 - eligibility change updates scenario-weighted replacement value.
-

8. Market architecture

8.1 Tag market source type

At minimum distinguish:

- crowdsourced stated preference;
- actual/revealed trades;
- expert consensus;
- startup ADP;
- rookie ADP;
- internal league market.

Source disagreement is not only noise. It can inform:

- market uncertainty;
- liquidity;
- format mismatch;
- narrative momentum;
- stale sources.

8.2 Normalize before blending

Normalize each source by:

- format;
- date;
- player universe;
- percentile or latent score;
- positional and pick anchors;
- known source scale.

Do not assume 5,000 means the same thing on every site.

8.3 Existing market integration

Audit:

- KTC;
- FantasyCalc;
- FantasyPros/DynastyProcess-style consensus;
- DLF or local CSV feeds;
- IDP Trade Calculator;
- any current ten-source consensus;
- league-adjusted values;
- the global valuation toggle.

Preserve useful existing ingestion. Replace duplicated business logic with canonical adapters.

Do not break existing calculator or ranking routes while BDVM is behind a feature flag.

8.4 Buy/hold/sell

A signal must consider:

- fundamental-market gap;
- fundamental confidence;
- market confidence;
- liquidity;
- trend;
- unresolved events;
- sensitivity to priors;
- strategic fit.

Do not flip recommendations because of minor weekly noise.

Store the prior signal and the reason it changed.

9. Rookie-pick integration

Represent a pick as a probability distribution, not a guaranteed player.

For pick q :

$$PickValue_q = E(PlayerValue\ selected) + LiquidityPremium + FlexibilityPremium - TimeDiscount + ClassStrengthAdjustment$$

Store:

- mean;
- median;
- P10/P90;
- starter probability;
- elite probability;
- bust probability;
- liquidity;
- time to selection;
- class-strength uncertainty.

Build from the platform's own historical rookie ADP and later player values where possible.

Use published hit rates only as sanity checks.

Support:

- 1QB;
- Superflex;
- TE premium;
- IDP-inclusive rookie drafts;
- pre-NFL Draft and post-NFL Draft states;
- current projected pick slot versus exact pick.

Starting strategy pick premiums may use the BDVM priors, but must remain configurable.

10. Roster Intelligence integration

The player model is neutral. Roster analysis is contextual.

For each roster calculate:

- current projected starter points;
- ROS optimized points;
- projected weekly best-ball points;
- one-, two-, three-, and five-year utility;
- fundamental asset value;
- market asset value;
- future pick value;
- contender capital;
- rebuild capital;
- now/future ratio;
- playoff probability;
- championship probability;

- competitive window;
- age concentration;
- risk concentration;
- liquidity;
- bench value above replacement;
- depth resilience;
- positional surplus;
- positional weakness;
- waiver replacement;
- future value cliffs.

For every position group calculate:

- rostered count;
- required slots;
- expected lineup entries;
- best-ball inclusion rate;
- elite assets;
- useful starters;
- depth assets;
- developmental assets;
- roster-clogger risk;
- age profile;
- injury exposure;
- weekly floor/ceiling;
- ROS value;
- future utility;
- market liquidity;
- replacement gap;
- league scarcity;
- tradeable surplus;
- target need.

Classify roster direction probabilistically:

- championship contender;
- playoff contender;
- fragile contender;
- fringe;
- retool;
- stuck in the middle;
- ascending;
- rebuild.

Allow user override. Store both model classification and user-selected strategy.

11. Trade Finder integration

11.1 The trade engine must answer

- Does the trade improve the user's roster?
- Does it plausibly improve the opponent?
- Is it externally market-fair?
- Is it favorable under the independent model?
- Does it fit both competitive windows?
- What is the best-player/consolidation effect?
- What is the roster-spot effect?
- Does it increase or reduce risk concentration?

- Does it improve playoff or championship odds?
- What assumptions could invalidate it?
- How likely is acceptance?

11.2 Candidate shapes

Support:

- one-for-one;
- two-for-one;
- one-for-two;
- two-for-two;
- three-for-two;
- player plus pick;
- offense plus IDP;
- pick-only packages where legal.

11.3 Search strategy

Avoid combinatorial explosion.

Prune using:

- market value band;
- fundamental value band;
- positional need;
- ownership;
- untouchables;
- trade block;
- strategy fit;
- roster legality;
- minimum opponent improvement;
- minimum user improvement;
- acceptance threshold;
- maximum package size.

Rank using Pareto logic across:

- user utility;
- opponent utility;
- market fairness;
- model edge;
- acceptance;
- championship impact;
- long-term health;
- risk;
- confidence.

11.4 Double-positive trades

A priority output is a trade that:

- the market views as sufficiently fair;
- improves the user according to roster-specific utility;
- is not harmful to the opponent's own roster utility;
- exploits differences in timeline, position need, or model-market disagreement.

Do not describe a trade as a “win” from one scalar alone.

12. User-facing outputs

12.1 Player card

Display:

- dynasty score;
- contender score;
- balanced score;
- rebuild score;
- ROS score;
- market score;
- market gap;
- market-adjusted score;
- trade units;
- rank by each view;
- floor;
- median;
- ceiling;
- confidence;
- volatility;
- probability above replacement;
- starter probability in Years 1 and 3;
- elite-asset probability;
- breakout probability;
- bust probability;
- one-year collapse probability;
- multi-year value curve;
- position and eligibility scenarios;
- contract and event markers;
- market history;
- model history.

12.2 Explanation

Every value must be decomposable.

Use module-level contribution or leave-one-component-out attribution.

Examples:

- current-year surplus share;
- Years 1–2 share;
- Years 3+ share;
- survival drag;
- trajectory effect;
- scarcity effect;
- role-security effect;
- uncertainty effect;
- contract effect;
- market disagreement;
- eligibility risk.

Generate three-sentence explanations without fabricating causal certainty.

12.3 Confidence policy

Do not publish precise user-facing probability claims until calibration is adequate.

Before calibration, labels may be:

- low;

- medium;
- high confidence;
- wide/narrow range;
- provisional.

Internally retain raw probabilities for validation.

13. Backtesting and model governance

13.1 Historical snapshots

A valid preseason historical row may use only information available by the prediction timestamp.

Prevent leakage from:

- final games played;
- future depth charts;
- end-of-season stats;
- revised past projections;
- future injury status;
- later contracts;
- later eligibility;
- current market values applied to the past;
- news published after the snapshot.

Enforce:

$$feature_timestamp \leq prediction_timestamp$$

13.2 Rolling-origin validation

Do not randomly shuffle player-seasons across time.

Use:

- expanding windows;
- rolling seasons;
- full-season holdouts;
- position-specific evaluation;
- experience-group evaluation;
- rookie versus veteran evaluation;
- IDP subtype evaluation.

13.3 Targets

Evaluate:

- next-season fantasy points;
- points per game;
- games;
- top-12/24/36 finishes;
- starter-level season;
- multi-year realized surplus;
- three- and five-year discounted value;
- startable-role retention;
- market appreciation;
- market decline;
- value retention;
- breakout;

- collapse.

13.4 Metrics

Use appropriate metrics:

- MAE;
- RMSE;
- Spearman;
- Kendall;
- NDCG@k;
- Brier score;
- log loss;
- CRPS;
- interval coverage;
- calibration slope and intercept;
- reliability curves;
- survival concordance;
- integrated Brier score;
- precision/recall and PR-AUC for breakout/decline;
- trade utility regret;
- actual recommendation acceptance.

13.5 Baselines

The full model must compare against:

1. prior-year points;
2. upcoming projection only;
3. projection plus simple age multiplier;
4. weighted three-year production;
5. market consensus;
6. market plus age;
7. plain $\mu - R$;
8. truncated $\max(0, \mu - R)$;
9. option value $E[\max(0, X - R)]$.

13.6 Ablations

Test removal of:

- age;
- survival;
- role ascension;
- contract;
- injury;
- draft capital;
- college;
- advanced opportunity;
- event regression;
- uncertainty;
- option surplus;
- market layer;
- archetype;
- eligibility scenarios.

Complexity must justify itself.

13.7 Champion–challenger

Production remains on a versioned champion.

A challenger must:

- train on past data only;
- beat the champion on multiple rolling folds;
- avoid material subgroup regressions;
- provide explanations;
- support rollback;
- never silently self-promote.

Production code and model logic must not autonomously rewrite themselves.

14. Implementation phases

Do not attempt to ship every module simultaneously.

Use the following dependency-first plan, adapting file names and duration to repository reality.

Phase 0 — Coordination, audit, and reference preservation

Objectives

- inspect active agents, branches, worktrees, and in-progress tasks;
- avoid duplicated work;
- identify the current valuation, projection, market, IDP, roster, trade, scoring, caching, and UI paths;
- preserve the BDVM reference artifacts;
- select the safest integration boundary.

Required actions

1. Locate `BDVM v1 complete.pdf`.
2. Extract Appendices A–F into an isolated reference directory, for example:
 - `docs/research/bdvm-v1/reference/`
 - or the repository’s equivalent non-production area.
3. Run `run_examples.py` unchanged beside `dynasty_engine.py`.
4. Confirm its output matches the embedded `examples_output.txt`.
5. Record the hash of the reference files.
6. Do not modify the reference files; create adapters or production modules elsewhere.
7. Audit:
 - projection sources;
 - player ID crosswalk;
 - exact scoring;
 - league config;
 - current replacement logic;
 - current trade value scales;
 - KTC;
 - IDP Trade Calculator;
 - consensus;
 - league-adjusted values;
 - ROS;
 - best ball;
 - Sleeper sync;
 - database;
 - scheduled jobs;
 - API contracts;
 - frontend consumers;

- tests;
 - caching;
 - feature flags.
8. Create an integration map showing:
 - retain;
 - adapt;
 - replace;
 - deprecate;
 - unknown.
 9. Begin daily market snapshots and weekly player-context snapshots immediately if not already archived.

Deliverables

- repository audit;
- BDVM reference fixture;
- architecture map;
- data-gap matrix;
- conflict list;
- exact first implementation slice;
- no unnecessary UI work.

Gate

Do not begin broad production rewrites until the current value flow and data contracts are mapped.

Phase 1 — Canonical schemas, league config, scoring, and projection adapters

Build

- canonical player identity;
- league config contract;
- scoring rules;
- lineup slots and eligibility;
- projection source adapters;
- raw stat scoring;
- projection snapshot storage;
- initial equal-weight/robust consensus;
- data quality and staleness checks;
- static/internal output contract.

Required data

- at least two credible projection sources where available;
- player IDs;
- league scoring;
- lineup rules.

Tests

- exact hand-scored examples;
- PPR/half/standard;
- TEP;
- six-point passing TD;
- multiple IDP formats;
- direct-fantasy-point fallback;
- stale/missing source behavior;
- raw stat and direct point consistency.

Gate

The same raw stat line must score correctly under multiple configs without changing model code.

Phase 2 — Dynamic replacement and upcoming-season surplus

Build

- flex-aware slot allocation;
- position groups;
- platform eligibility;
- waiver buffer;
- rank-based VORP;
- expected positive surplus;
- option-value formula;
- VOLS diagnostic;
- league-neutral current-year ranks.

Tests

- five manually verified players per position;
- monotonicity by team count;
- monotonicity by required starters;
- Superflex behavior;
- TEP behavior;
- DT-required versus combined DL;
- CB/S-required versus combined DB;
- LB versus safety scarcity;
- option-value ablation.

Gate

No cross-position rank is accepted until replacement logic passes mechanical and football sanity checks.

Phase 3 — Aging, mileage, ascension, survival, games, and strategy

Build

- position/archetype curves;
- effective age;
- role ascension;
- draft and college decay;
- hazard model;
- expected games;
- multi-year path;
- contender/balanced/rebuilder profiles;
- risk-neutral output;
- floor/median/ceiling;
- explanation decomposition.

Tests

- no double-counting age in curve and hazard;
- S(3) calibration;
- old RB versus old QB;
- young low-projection player does not automatically outrank a proven star;
- young TE ascension;

- dual versus pocket QB;
- career-load sensitivity;
- risk-profile sensitivity.

Gate

Every future-year value must be explainable as current mean $\times$ trajectory $\times$ ascension $\times$ games $\times$ survival $\times$ scarcity $\times$ strategy.

Phase 4 — IDP first-class production models

Build

- true position;
- platform position;
- configurable DL/DT/EDGE;
- configurable DB/CB/S;
- pass-rush and expected sack logic;
- tackle opportunity;
- LB role;
- DB alignment;
- CB-specific model;
- eligibility scenarios;
- IDP confidence fallback.

Data tiers

- Tier A: proprietary alignment/pressure data;
- Tier B: public snap/PBP proxies;
- Tier C: projection-only fallback.

Output confidence must reflect the tier.

Tests

- DT format sensitivity;
- EDGE sack regression;
- full-time LB versus two-down LB;
- box safety versus deep safety;
- volatile CB;
- eligibility flip;
- combined versus true-position formats.

Gate

IDP values cannot be published as equal-confidence outputs when essential role data are unavailable.

Phase 5 — Market comparison and normalization

Build

- source type tags;
- daily snapshots;
- latent normalization;
- offense-IDP cross-market mapping;
- dispersion;
- liquidity;

- market gap;
- alpha;
- momentum;
- market-adjusted output;
- buy/hold/sell.

Tests

- fundamental model can run with market tables unavailable;
- market cannot alter stored fundamental value;
- raw incompatible values are never directly summed;
- normalization version is retained;
- stale source fallback;
- mixed offense-IDP trade fairness.

Gate

The fundamental output must be fully reproducible without market data.

Phase 6 — Contract, injury, role, and news events

Build

- contract ingest;
- practical exit year;
- guarantee and dead-cap signals;
- structured event ontology;
- event reflection detection;
- confidence and decay;
- role competition;
- coordinator-change stability flags;
- injury-specific availability.

Validation

Replay known historical events using only information known at that time.

Gate

No arbitrary manual “+10%” adjustments to final values.

Phase 7 — Roster Intelligence

Build

- Sleeper or current league-platform sync;
- roster snapshots;
- optimized lineup and best-ball utility;
- positional profiles;
- competitive-window forecast;
- surplus/deficit;
- age/risk/liquidity concentration;
- contender/retool/rebuild recommendation;
- user strategy override;
- explanations.

Validation

- expert review of at least 20 real rosters;
- lineup legality;
- marginal utility checks;
- strategy stability;
- IDP included throughout.

Gate

Roster recommendations must use actual marginal utility rather than only summed trade values.

Phase 8 — Trade finder and partner engine

Build

- opponent profiles;
- candidate generation;
- CES display math;
- roster-spot costs;
- before/after roster utility;
- market fairness;
- model edge;
- double-positive filtering;
- acceptance estimate;
- multi-asset search;
- Pareto ranking;
- recommendation explanation.

Validation

- all supported trade shapes;
- no duplicate or unowned assets;
- legal post-trade rosters;
- both roster utilities;
- mixed-market normalization;
- recommendation acceptance and rejection reasons;
- runtime limits.

Gate

A trade cannot be labeled recommended without a complete versioned evaluation for both sides.

Phase 9 — Rookie-pick distributions

Build

- pick-slot distributions;
- league-format splits;
- class strength;
- liquidity premium;
- flexibility premium;
- time discount;
- expected/median/ceiling/bust;
- pre- and post-NFL Draft updates.

Validation

- historical calibration;
 - slot monotonicity where appropriate;
 - honest uncertainty;
 - no guaranteed “player equivalent.”
-

Phase 10 — ROS and season simulation

Build

- remaining schedule;
- bye weeks;
- weekly active probability;
- expected return;
- ramp;
- current usage;
- playoff weighting;
- trade deadline;
- optimized roster baseline;
- weekly best-ball;
- playoff/championship simulations.

Validation

- weekly projection error;
 - lineup inclusion;
 - playoff probability calibration;
 - no future-week information leakage.
-

Phase 11 — Historical backtesting and calibration

Build

- rolling datasets;
- baselines;
- ablations;
- calibration;
- subgroup reports;
- value-retention tests;
- model-versus-market tests;
- published internal model card.

Gate

Do not claim the priors are validated until this phase demonstrates it.

Phase 12 — Targeted ML challengers

Only after sufficient clean snapshots exist.

Candidates:

- role ascension;
- games missed;
- starter-loss hazard;

- projection blending;
- trade acceptance;
- market movement.

Do not replace the transparent skeleton unless the challenger wins under the governance rules.

15. Repository and agent coordination rules

- Inspect existing agents and autonomous workflows before assigning work.
 - Do not duplicate another agent's active module.
 - Prefer shared schemas and stable interfaces before parallel implementation.
 - Use a safe branch/worktree if the repository workflow requires it.
 - Avoid unnecessary pull-request and merge churn.
 - Preserve other agents' work.
 - Do not rewrite broad areas of the site merely to fit the reference code.
 - Keep the current production path stable through adapters and feature flags.
 - Do not spend early phases polishing UI while the value contracts remain unstable.
 - Do not claim a phase is complete if it is only documented.
 - Do not claim implementation without tests and actual file changes.
 - Do not let a failed refresh replace a last-known-good snapshot.
 - Do not let the model autonomously modify production code or promote itself.
 - Respect the existing global league-adjusted valuation toggle. Do not create a conflicting local toggle.
-

16. Stable service contracts

Adapt to the project language and architecture, but aim for clear interfaces equivalent to:

```

ingestProjectionSnapshot(...)
buildProjectionConsensus(...)
scoreProjection(...)
resolveEligibility(...)
calculateReplacementLevels(...)
calculateCurrentSeasonSurplus(...)
calculateTrajectory(...)
calculateRoleAscension(...)
calculateSurvivalPath(...)
calculateGamesPath(...)
calculateDynastyDistribution(...)
calculateStrategyValues(...)
calculateMarketConsensus(...)
normalizeMarketValues(...)
calculateMarketGap(...)
calculateRestOfSeasonValue(...)
calculatePickDistribution(...)
analyzeRoster(...)
classifyCompetitiveWindow(...)
calculatePositionStrengths(...)
profileTradePartner(...)
generateTradeCandidates(...)
evaluateTradePackage(...)
estimateTradeAcceptance(...)
rankTradeRecommendations(...)
explainPlayerValue(...)

```

```
explainTradeRecommendation(...)
publishValuationSnapshot(...)
```

Keep algorithms behind stable contracts so the parametric champion can later be replaced module by module.

17. API output contract

A player output should support an equivalent shape:

```
{
  "player_id": "canonical-id",
  "as_of": "timestamp",
  "league_config_id": "config-id",
  "model_version": "version",
  "parameter_set_id": "params",
  "position": {
    "true_position": "EDGE",
    "platform_eligibility": ["DL", "LB"],
    "eligibility_scenarios": []
  },
  "projection": {
    "fpg": 0,
    "games": 0,
    "source_dispersion": 0,
    "source_count": 0,
    "confidence": 0
  },
  "replacement": {
    "group": "DL",
    "fpg": 0,
    "method": "dynamic_vorp",
    "run_id": "replacement-run"
  },
  "values": {
    "fundamental_expected": 0,
    "contender": 0,
    "balanced": 0,
    "rebuilder": 0,
    "risk_neutral": 0,
    "rest_of_season": 0,
    "market": 0,
    "market_adjusted": 0,
    "market_gap": 0,
    "market_alpha": 0,
    "trade_units": 0,
    "dynasty_score_0_100": 0
  },
  "distribution": {
    "floor_p10": 0,
    "median_p50": 0,
    "ceiling_p90": 0,
    "volatility": 0,
    "confidence": 0
  },
  "probabilities": {
    "above_replacement": 0,
```

```

    "starter_y1": 0,
    "starter_y3": 0,
    "elite_asset": 0,
    "breakout": 0,
    "bust": 0,
    "collapse_1y": 0
  },
  "path": [],
  "market": {
    "sources": [],
    "dispersion": 0,
    "liquidity": 0,
    "normalization_version": "version",
    "confidence": 0
  },
  "explanation": {
    "drivers": [],
    "current_year_share": 0,
    "years_1_2_share": 0,
    "years_3_plus_share": 0,
    "survival_drag": 0,
    "risk_drivers": [],
    "missing_data": []
  },
  "quality": {
    "status": "valid|provisional|stale|failed",
    "imputed": false,
    "warnings": []
  }
}

```

Do not force the repository to use this exact JSON if an existing canonical contract is better. Preserve equivalent semantics.

18. Testing requirements

Add tests for:

Projection and scoring

- source blending;
- source staleness;
- raw-stat scoring;
- direct-point fallback;
- all supported scoring families;
- TEP;
- IDP scoring;
- missing sources.

Replacement

- flex allocation;
- Superflex;
- TE premium;
- IDP flex;
- DT-required;

- CB/S-required;
- roster depth;
- waiver buffer;
- monotonicity;
- multi-position eligibility.

Dynasty math

- option surplus;
- plain surplus ablation;
- trajectory;
- mileage;
- draft decay;
- college decay;
- survival;
- expected games;
- strategy horizons;
- risk-neutral output;
- deterministic reproducibility.

IDP

- expected sack regression;
- LB role;
- CB target dependence;
- box/deep safety;
- designation scenarios;
- confidence tiers.

Market

- fundamental isolation;
- source normalization;
- mixed offense-IDP;
- market gap;
- stale source;
- liquidity;
- momentum guards.

Roster/trade

- legal lineup;
- best-ball;
- positional need;
- competitive window;
- one-for-one;
- multi-asset;
- roster-spot cost;
- both-roster utility;
- ownership;
- no duplicate assets;
- acceptance;
- explanation;
- recommendation stability.

Operations

- as-of leakage;

- snapshot immutability;
 - model versioning;
 - atomic publication;
 - stale fallback;
 - rollback;
 - cache-key compatibility;
 - feature flag;
 - performance;
 - accessibility for user-facing additions.
-

19. Acceptance criteria

The integrated Version 1 is not complete unless:

1. Every rostered offensive and defensive player can receive a fundamental value under a supported league config.
 2. IDP is integrated through the same core pipeline.
 3. Fundamental value is computable with no market data.
 4. Current projection context is not double-counted.
 5. Aging and survival are separate.
 6. Role ascension exists.
 7. Replacement is dynamic and flex-aware.
 8. Position eligibility is versioned and scenario-capable.
 9. Contender, balanced, and rebuilder values are generated inside the horizon calculation.
 10. Risk-neutral value is available.
 11. Market value, market gap, and market-adjusted value remain distinct.
 12. Raw incompatible offense and IDP market values are never directly summed.
 13. Rest-of-season value remains separate.
 14. Roster-specific value uses actual marginal utility.
 15. Trade evaluation considers both rosters.
 16. Rookie picks are distributions.
 17. Every output is versioned and timestamped.
 18. Missing/stale data cannot silently produce a normal-looking value.
 19. Every player card can explain the main value drivers.
 20. Backtest infrastructure prevents look-ahead leakage.
 21. The option-value formula is tested against simpler baselines.
 22. Production promotion uses champion–challenger governance.
 23. Existing production features are not broken during migration.
 24. No feature is called complete unless implemented and tested.
-

20. Required implementation report

Maintain a living report with:

1. Executive summary
2. Repository findings
3. Existing valuation architecture
4. Existing market architecture
5. Existing IDP architecture
6. Existing roster/trade architecture
7. Agent coordination
8. BDVM reference extraction and verification
9. Retain/adapt/replace/deprecate map
10. Data availability and missing-data tiers
11. Schema and migrations

12. Services and interfaces
13. Projection ingestion
14. Replacement methodology
15. Dynasty methodology
16. IDP methodology
17. Market normalization
18. Roster methodology
19. Trade methodology
20. Backtesting and leakage controls
21. Jobs and cadence
22. Feature flags and migration
23. Files changed
24. Tests run and results
25. Performance findings
26. Missing credentials or proprietary data
27. Risks and limitations
28. Material judgment calls
29. Remaining work
30. Rollback plan
31. Exact next actions

Be explicit about what is:

- implemented;
- reference-only;
- scaffolded;
- provisional;
- blocked;
- awaiting data;
- awaiting validation.

21. Immediate first actions

Begin now with concrete repository work.

Your first actions must be:

1. Inspect repository status, branches/worktrees, active agents, and current changes.
2. Locate the live valuation, rankings, scoring, projection, market, IDP, roster, trade, database, API, cache, and UI paths.
3. Locate **BDVM v1 complete.pdf**.
4. Extract and preserve Appendices A–F in a non-production reference directory.
5. Run the reference examples and verify the output.
6. Map the reference model to existing production components.
7. Audit current KTC, IDP Trade Calculator, FantasyCalc, consensus, and league-adjusted normalization.
8. Audit whether daily market and weekly player-context snapshots already exist.
9. Start or specify the snapshot jobs immediately because missing historical data cannot be reconstructed later.
10. Identify the safest Phase 1 implementation slice.
11. Create canonical contracts or migrations needed for league config, projection snapshots, and model versioning.
12. Begin implementing the Phase 1 scoring/projection/replacement foundation behind a feature flag.
13. Run tests.
14. Report actual files changed, actual commands/tests run, blockers, and the next implementation slice.

Do not respond with only a proposal.

Do not spend the response repeating this prompt.

Do not wait for approval for every minor decision.

Make the safest concrete progress possible, and clearly surface only the decisions that genuinely require user input.

22. Final decision rule

When uncertain, use this rule:

- preserve the independent fundamental model;
- preserve current production behavior until the replacement path is tested;
- choose the simplest transparent method that passes out-of-time validation;
- keep uncertain priors configurable;
- expose uncertainty rather than hiding it;
- use roster marginal utility for personalized decisions;
- keep market price separate from model worth;
- leave advanced ML for targeted challengers;
- document any unresolved issue without allowing it to block the implementable baseline.

The desired end state is not merely another ranking table.

It is a versioned dynasty intelligence platform where upcoming-season projections become league-specific, cross-position, multi-year probability distributions; those distributions become independent player values; player values are compared to the market; and actual roster context determines which trades, targets, and strategic actions improve a team.